

The Role of Mathematics in Machine Learning

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Leads to:** How to learn the maths for ML ([Note MA-002](#)).

1. Overview

Key point: Four branches of mathematics carry most of ML: linear algebra stores the data, calculus improves the model, probability decides when the data is unclear, and statistics draws conclusions from the data.

ML is not magic: every algorithm is mathematics running on data. To understand why an algorithm works, and to fix it when it does not, we need the mathematics underneath.

Four branches of maths in ML

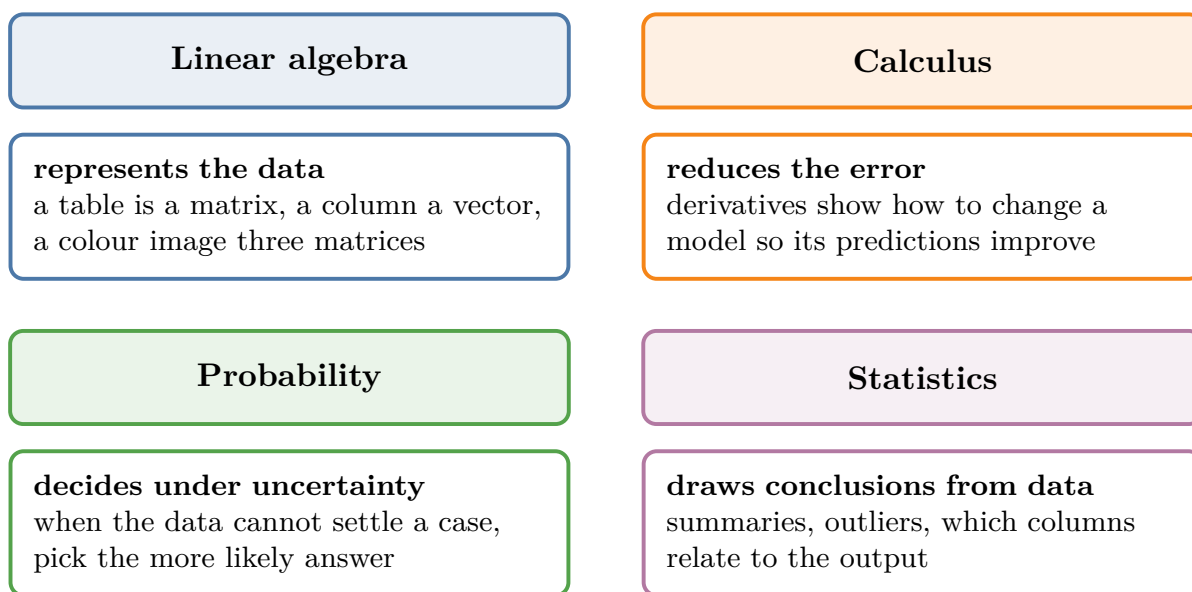


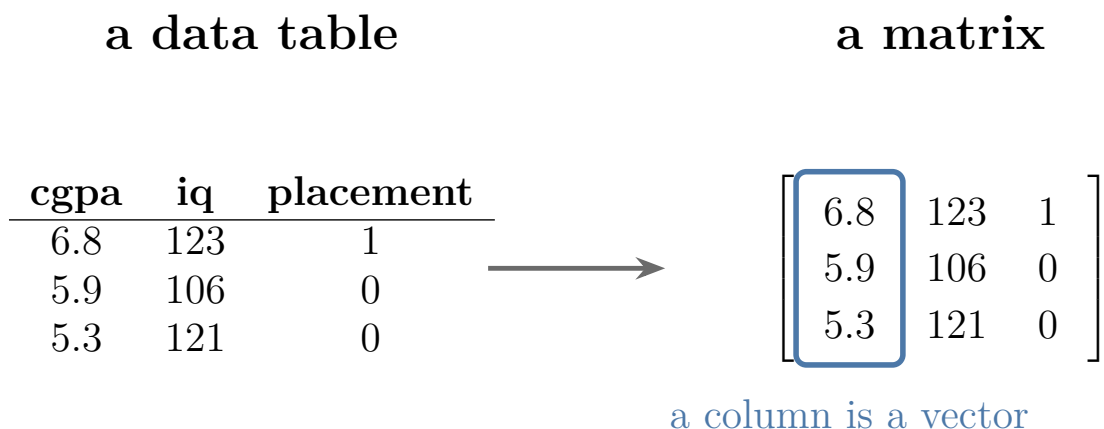
Figure 1: The four branches of mathematics used in ML, and the job each one does

Figure 1 shows the four branches and the job each one does. This Note gives a first feel for each job and points to the Notes where each branch is taught. The details come in those Notes.

2. Linear algebra: representing data

Key point: Linear algebra gives us the containers for data (vectors, matrices and tensors) and the operations that act on a whole container at once.

Linear algebra (G-1090) stores every kind of data (tables, text, images, video) as vectors (lists of numbers, such as $(2, 3)$), **matrices** (G-1180; tables of numbers) and **tensors** (G-1957; grids of numbers with any number of axes), and acts on a whole container in one step; [why ML needs linear algebra](#) explains why. How each kind of data becomes a tensor is shown in [tensors from 0D to 5D](#).



first three rows of the placement data (toy project Note)

Figure 2: Three rows of the placement data as a table and as a matrix. Each column of the matrix is a vector

Figure 2 shows the first step of linear algebra in ML: the table we read becomes a matrix the computer can act on in one step.

[The eight modules of the linear algebra roadmap](#) list every linear algebra topic ML needs.

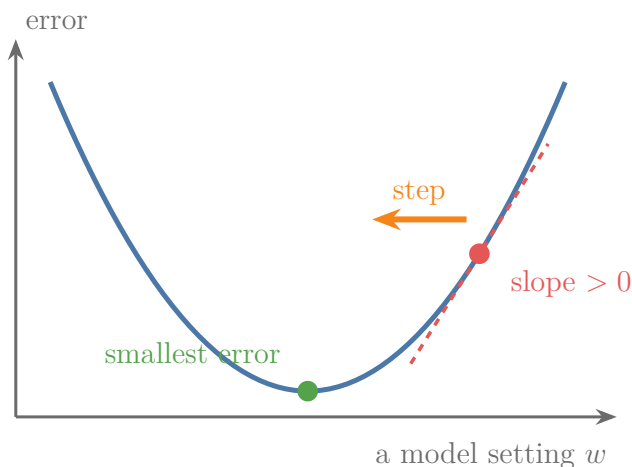
3. Calculus: reducing the error

Key point: Predictions are never perfect; calculus tells us how to change a model so that its error shrinks.

Calculus (G-339) is the branch of mathematics about change: differentiation (slopes) and integration (areas). For beginners it is usually the hardest of the four.

An ML model predicts something, and its predictions are never exactly right. The gap between the prediction and the truth is the **error**. Training a model means changing it until the error is as small as we can get it, which is called **optimisation** (G-1400).

Most optimisation methods are built on calculus. The **derivative** (G-595) of the error tells us in which direction to change each part of the model so the error goes down. **Gradient descent** (G-862) does exactly this, one small step at a time.



the slope (derivative) says which way is uphill;
step the other way

Figure 3: Error against one model setting w . At the red point the slope is positive, so a step to the left lowers the error

Figure 3 shows the idea: the sign of the slope says which way the error rises, and the model steps the other way, towards the green minimum. Figure 4 repeats the step until the slope is flat: each step is the slope times a small number (here 0.3), so the steps shrink as the model nears the minimum. Here w is one setting of the model, such as the slope of a line it fits, and the error curve is:

$$E(w) = (w - 2)^2 + 1$$

Starting at $w = 5$, the slope of the error curve is $2(w - 2)$, and each row below is one step (new $w = \text{old } w \text{ minus } 0.3 \times \text{slope}$):

Step	w	Slope	Error
0	5	6	10
1	3.2	2.4	2.44
2	2.48	0.96	1.23
3	2.19	0.38	1.04
4	2.08	0.15	1.01
5	2.03	0.06	1.00
6	2.01	0.02	1.00
7	2.00	0.01	1.00

Step	w	Slope	Error
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For example, step 1 moves w from 5 against the slope 6 (step size 0.3):

$$5 - 0.3 \times 6 = 3.2$$

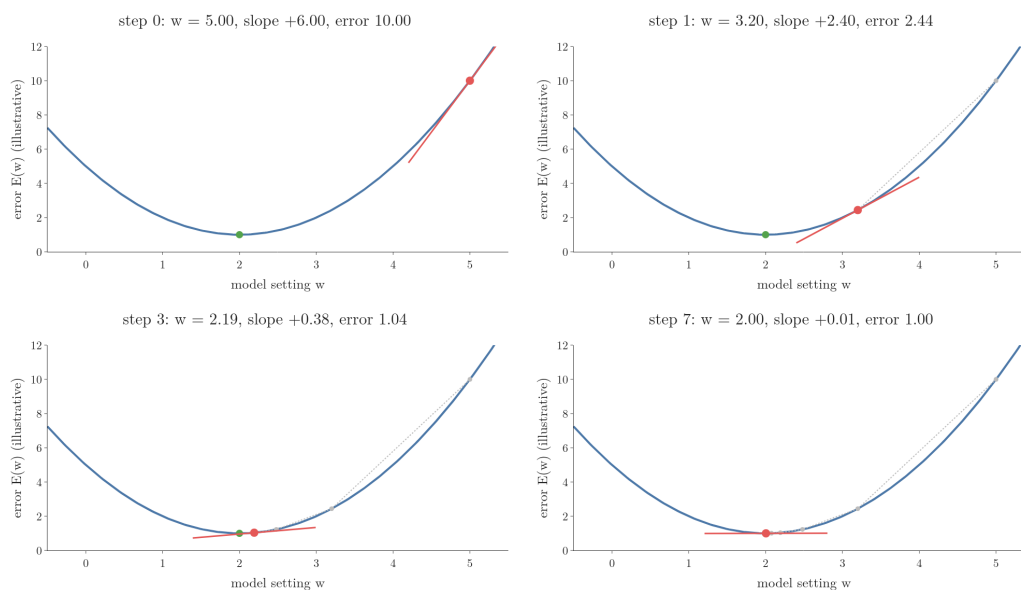


Figure 4: Gradient descent on an illustrative error curve $E(w) = (w - 2)^2 + 1$, from $w = 5$ with step size 0.3. The red line is the slope at the current point; the error falls from 10 to 1.00 in seven steps.

Extra: A model with 100% accuracy is a warning sign, not a success. Real outputs contain a random part that no model can predict, called the **irreducible error** (G-1327; ISLR §2.1.1). So a perfect score often signals a mistake, such as test data leaking into training (**data leakage** (G-535), see [scaling the inputs](#); Kaufman et al. 2012).

Calculus first appears in [finding the minimum](#), where a derivative set to zero gives the best line. Derivatives themselves are taught step by step in [from secant to tangent](#).

4. Probability: deciding in the grey zone

Key point: When the data cannot settle a case for certain, probability tells us which answer is more likely, and we choose that one.

Often the data leaves a grey zone, where no answer is certain. Suppose we want to guess a person's gender from their salary alone. In the salary range 4 to 5 lakh rupees, there are both men and women, so salary cannot decide.

Probability (G-1574) still lets us decide. If 40% of the people in that range are men and 60% are women, then for a new person earning 4.5 lakh rupees the more likely answer is “woman”, and that is what we predict. The chances inside one range add up to 100%.

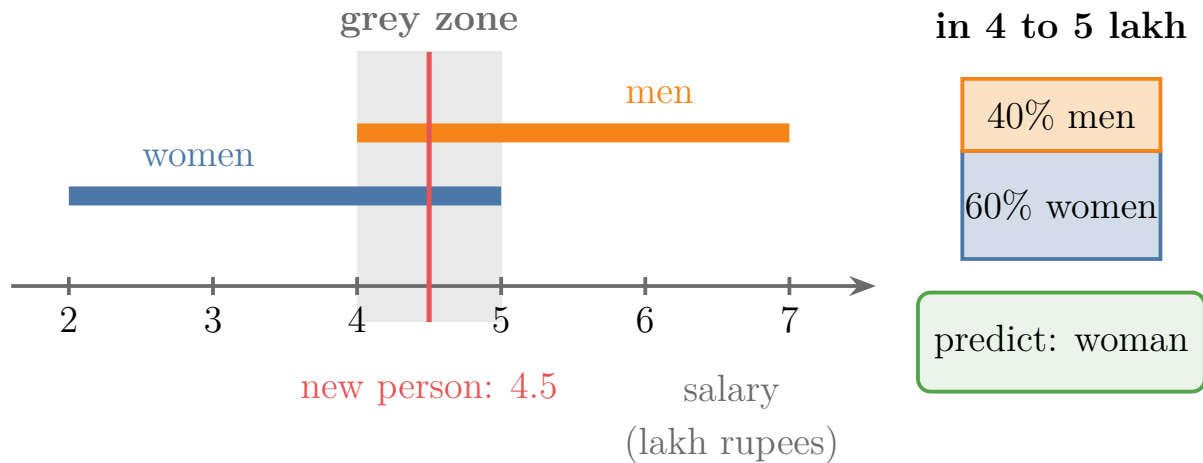


Figure 5: Salary cannot separate the two groups between 4 and 5 lakh rupees (the ranges outside that zone are drawn for illustration). With the shares 60% women and 40% men in that range, the prediction for a new person earning 4.5 lakh is “woman”

Figure 5 draws the grey zone: both groups overlap there, so we predict the class with the larger share.

Picking the most probable class is how probabilistic classifiers such as [Naive Bayes](#) (G-1297) work: compute the probability of every class, then pick the largest. The rules of probability are taught from [the five basic terms of random experiments and events](#) to [joint, marginal and conditional probability](#), and [Bayes’ theorem](#) builds on them.

5. Statistics: drawing conclusions from data

Key point: Statistics is used the most of the four: from the moment we receive data to the moment we use the result.

Statistics (G-1884) is the branch of mathematics for collecting and analysing data so that we can draw conclusions from it (see [what statistics is](#)). It appears at every stage of an ML project.

In real work, nobody hands us a dataset with a list of questions. We get the data and must find the meaningful questions ourselves, then answer them. Statistics supplies the tools:

- **Is there an outlier?** One value far from all the others can mislead a model (see [how outliers spoil a model](#)).
- **Is the data noisy?** Statistics helps us spot noisy data.

- **Which algorithm suits this data?** The shape of the data points to suitable algorithms.
- **Is this feature related to the target?** A **feature** (G-772) is an input variable (one column of the data table), and the **target** (G-1949) is the output we predict. If a feature is unrelated to the target, there is no reason to feed it to the algorithm. Measures such as [correlation](#) answer this.

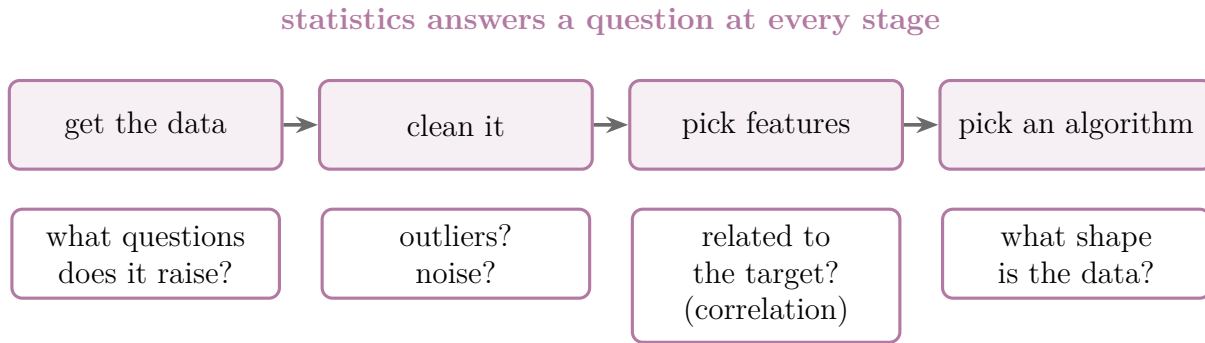


Figure 6: The questions statistics answers at each stage of an ML project

Figure 6 places the four questions along a project, from receiving the data to choosing the algorithm.

Data analysis, which ML depends on heavily, is built almost entirely on statistics (see [what the data looks like in numbers](#)). [The four modules of the statistics roadmap](#) map the whole subject.

6. Summary

Branch	Its job in ML	Example	Where it is taught
Linear algebra	Represent data and act on it	A table as a matrix	Vectors , tensors
Calculus	Reduce the error (optimisation)	Gradient descent	Gradient descent
Probability	Decide under uncertainty	Pick the more likely class	Events onward
Statistics	Draw conclusions from data	Find outliers, related features	Statistics roadmap

- ML is mathematics running on data; four branches carry most of it.
- Statistics is used the most; linear algebra does the heavy lifting of storing and transforming data.

7. Sources

Built from

- CampusX, “Role of Mathematics in Machine Learning”, YouTube, <https://www.youtube.com/watch?v=>

Other references

- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer. Section 2.1.1, reducible and irreducible error.
- Kaufman, S., Rosset, S. and Perlich, C. (2012). “Leakage in Data Mining: Formulation, Detection, and Avoidance”. *ACM Transactions on Knowledge Discovery from Data* 6(4).

8. Key terms

Term	Meaning
Calculus	The branch of mathematics about change: differentiation and integration
Error (G-705)	The gap between a model’s prediction and the true value
Optimisation	Changing a model step by step until its error is as small as possible